PROJECTILES

Physics Revision Cheatsheet • Senior Secondary • WAEC / NECO / JAMB

PrepNAce

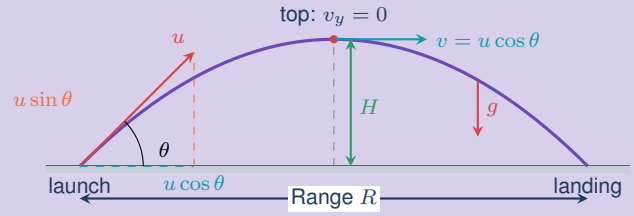
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1 What is a Projectile?

A **projectile** is any object thrown or launched into the air that moves **under gravity alone** (air resistance ignored). Examples: kicked football, thrown stone, water from a hose, cannon ball.

- Path (trajectory) is a **parabola**.
- **Horizontal** and **vertical** motions are **independent**.
- Horizontal: **constant velocity** ($a_x = 0$).
- Vertical: **constant acceleration** $g = 10$ m/s^2 downward (use 9.8 if told).

2 Trajectory Diagram



3 Resolving the Launch Velocity

Launch speed u at angle θ **above the horizontal**:

$$\boxed{u_x = u\cos\theta} \qquad \boxed{u_y = u\sin\theta}$$

- Angle given from the **vertical**? Swap $\sin \leftrightarrow \cos$.
- Speed at any time: $v = \sqrt{v_x^2 + v_y^2}$
- Direction: $\tan\alpha = \dfrac{v_y}{v_x}$ (below horizontal once $v_y < 0$)

4 Equations of Motion

	Horizontal	Vertical
Accel.	$a_x = 0$	$a_y = -g$
Velocity	$v_x = u\cos\theta$	$v_y = u\sin\theta - gt$
Position	$x = u\cos\theta\, t$	$y = u\sin\theta\, t - \frac{1}{2}gt^2$
Extra	—	$v_y^2 = u^2\sin^2\theta - 2gy$

Key Formulas (launch & land at same level)

Time of flight	$T = \dfrac{2u\sin\theta}{g}$
Time to top	$t_{\text{top}} = \dfrac{u\sin\theta}{g} = \dfrac{T}{2}$
Max height	$H = \dfrac{u^2\sin^2\theta}{2g}$
Range	$R = \dfrac{u^2\sin 2\theta}{g}$

5 At the Top & Special Angles

- At the top: $v_y = 0$, but speed $= u\cos\theta$ (**not zero**, it is the minimum speed). Acceleration is still g downward.
- $45°$ gives **maximum range**: $R_{\max} = \dfrac{u^2}{g}$.
- **Complementary angles** (θ and $90° - \theta$) give the **same range**.
- $90°$ (straight up): max height $\dfrac{u^2}{2g}$, range $= 0$.
- $R \propto u^2$, $H \propto u^2$, $T \propto u$: **double** $u \Rightarrow R$ and H become $\times 4$, T becomes $\times 2$.
- Mass does **not** affect the motion.

6 Horizontal Launch (from height h)

Thrown horizontally at speed u from height h: $u_y = 0$.

$$t = \sqrt{\frac{2h}{g}}, \qquad R = u\,t, \qquad v_y = gt, \qquad v = \sqrt{u^2 + 2gh}$$

7 Worked Examples

Ex 1. A ball is kicked at 20 m/s at $30°$ ($g = 10$). Find T, H, R.

$T = \dfrac{2(20)\sin 30°}{10} = $ **2 s** $\quad H = \dfrac{20^2\sin^2 30°}{2(10)} = $ **5 m**

$R = \dfrac{20^2\sin 60°}{10} = 40(0.866) \approx$ **34.6 m**

Ex 2. A stone is thrown horizontally at 15 m/s from a 45 m cliff ($g = 10$). Find the time and horizontal distance.

$t = \sqrt{\dfrac{2(45)}{10}} = $ **3 s** $\qquad R = 15 \times 3 =$ **45 m**

8 Exam Tips & Common Mistakes

- ✓ Split the problem: **x** uses constant speed, **y** uses g.
- ✓ Time t is the **link** between x and y.
- ✓ Pick a sign rule (up $+$, down $-$) and keep it.
- × Don't say velocity is zero at the top, only $v_y = 0$.
- × Don't use $\cos\theta$ for the vertical part.
- × R and H formulas apply only if landing level equals launch level.